REVANTH G

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SUMMARY

- Senior Data Engineer with 9+ years of experience delivering enterprise-scale data solutions across healthcare, insurance, telecom, and banking domains.
- Proven expertise in architecting and developing data pipelines using modern cloud platforms including Azure (ADF, Synapse, Databricks), AWS (Glue, Kinesis, SageMaker), and Snowflake.
- Skilled in building scalable ETL/ELT workflows, streaming data architectures, and lakehouse platforms using PySpark, Delta Lake, and real-time ingestion tools (Kafka, Flink, Event Hubs).
- Strong background in data modeling (Star, Snowflake, Data Vault), performance optimization, and schema evolution across structured and semi-structured data sources.
- Hands-on experience with key orchestration and workflow tools like Airflow, Azure Data Factory, and AWS Step Functions to automate and manage complex pipelines.
- Implemented robust data governance, access control, and compliance (HIPAA, SOC 2, GDPR) using tools like Azure AD, IAM, and Key Vault.
- Adept at CI/CD integration using GitHub Actions, Terraform, and Jenkins for deploying and maintaining cloud-native data solutions.
- Versatile with business intelligence tools (Power BI, Tableau) for building executive dashboards and enabling self-service analytics.
- Recognized for driving cross-functional collaboration, mentoring junior engineers, and leading cloud migration and modernization initiatives.

TECHNICAL SKILLS

Big Data & Analytics	Apache Spark (Core, SQL, MLlib, Streaming), PySpark, Databricks, Delta Lake, Apache Flink, Hive, HDFS
Cloud Platforms (Azure)	Data Factory, Synapse Analytics, Databricks, Event Hubs, Functions, ADLS Gen2, Key Vault, Monitor
Cloud Platforms (AWS)	Glue, Kinesis, SageMaker, Athena, Lambda, S3, IAM, CloudWatch, EMR
ETL & Orchestration	Apache Airflow, Azure Data Factory, AWS Glue, SSIS, Informatica PowerCenter, dbt
Data Warehousing & Storage	Snowflake, Azure Synapse Analytics, Amazon Redshift, SQL Server, Oracle, PostgreSQL
Streaming & Messaging	Kafka (familiar), AWS Kinesis, Azure Event Hubs, Apache Flink
Programming & Scripting	Python, SQL, Scala, Shell Scripting, PowerShell
DevOps & IaC	Git, GitHub Actions, Jenkins, Terraform, CI/CD Pipelines, Azure DevOps
Data Governance & Security	Role-Based Access Control (RBAC), Azure Active Directory, AWS IAM, Key Vault, KMS Encryption, HIPAA, GDPR, SOC 2
Data Visualization & BI	Power BI, Tableau, SSRS, KPI Dashboards, Executive Reporting
File Formats & Tools	Parquet, Avro, JSON, CSV, XML; Great Expectations, Postman
Methodologies & Practices	Agile (Scrum), Test-Driven Development (TDD), Data Lakehouse, Data Vault 2.0, Confluence, JIRA
Certifications	Microsoft Azure Data Engineer Associate, Snowflake Snowpro Core

EXPERIENCE

Optum **Remote, USA**
Senior Data Engineer **Jul 2022 – Present**

- Designed and built a population health analytics platform using Azure Synapse Analytics, Data Factory, and ADLS Gen2 to support value-based care programs.
- Developed scalable ETL pipelines in Azure Data Factory for ingesting claims, EHR, and member data from on-prem and cloud sources into curated zones.
- Implemented real-time data ingestion via Azure Event Hubs and processed clinical events using Azure Stream Analytics for downstream analytics.
- Leveraged Delta Lake on Azure Databricks for incremental ingestion and version-controlled transformations.
- Integrated PySpark jobs into ADF pipelines for large-scale data transformation and enrichment.
- Developed role-based access controls (RBAC) using Azure Active Directory and enforced compliance with HIPAA and internal data governance policies.
- Applied encryption strategies using Azure Key Vault for secrets management and Azure Monitor for observability and incident tracking.
- Built Power BI dashboards to visualize patient cohort metrics, cost drivers, and provider performance KPIs.
- Deployed GitHub Actions for CI/CD of ADF pipelines and Databricks notebooks.
- Enabled lakehouse architecture by unifying structured and semi-structured data in Delta format for scalable analytics.
- Conducted root cause analysis on failed pipelines using ADF diagnostics and Spark logs, improving SLA compliance by 25%.
- Orchestrated data ingestion from external APIs using Azure Functions and Logic Apps to enrich clinical datasets.
- Tuned Spark configurations in Synapse Spark pools to optimize memory usage for 1B+ record datasets.
- Established reusable templates and parameterized pipelines for onboarding new data sources quickly.
- Created technical documentation and confluence wikis for pipeline architecture, metadata contracts, and data quality rules.
- Partnered with product owners, compliance, and security teams to implement audit logging and ensure GDPR alignment.
- Mentored junior engineers and led knowledge-sharing sessions on Delta Lake best practices and Azure resource management.
- Facilitated cross-functional planning with BI and Data Science teams for scalable dataset design.
- Migrated legacy SQL-based pipelines to modular PySpark frameworks to support long-term maintainability and performance.
- Implemented monitoring via Azure Monitor and Log Analytics to track pipeline latency, error rates, and throughput.
- Integrated Snowflake as a secondary analytics layer for cross-cloud reporting and ad hoc query workloads using federated queries.

Environment: Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Delta Lake, Azure Event Hubs, Azure Stream Analytics, Azure Functions, Azure Key Vault, ADLS Gen2, Power BI, GitHub Actions, Python, PySpark, SQL, REST APIs, Azure Monitor, Logic Apps, Azure Active Directory, Confluence, JIRA, HIPAA, GDPR

CSAA Insurance **TX, USA**
Senior Data Engineer **Jan 2020 – Jun 2022**

- Engineered a telematics ingestion pipeline using AWS Kinesis, Lambda, and Apache Flink to enable real-time pricing models.
- Integrated driving behavior analytics into actuarial models, increasing personalization and customer retention.
- Developed and deployed fraud detection models using scikit-learn and AWS SageMaker, improving claim fraud detection accuracy.
- Built ETL workflows in AWS Glue for claims, policies, and payment datasets using Spark and Python.
- Managed metadata and schema evolution using AWS Glue Data Catalog and JSON schemas.
- Implemented role-based access using IAM policies and encrypted sensitive data with AWS KMS.
- Queried terabytes of data using Amazon Athena for ad hoc investigations and pipeline validations.
- Designed and tested ML feature pipelines and model tracking using TDD and CI/CD principles.
- Tuned Spark jobs on EMR clusters to reduce processing latency and cost.
- Automated infrastructure provisioning via Terraform to deploy reusable pipeline environments.
- Integrated pipeline monitoring with AWS CloudWatch and set alarms for anomaly detection.
- Led the migration from batch S3 ingestion to real-time streaming for telematics workflows.
- Collaborated with claims and data science teams to enhance model explainability and interpretability.
- Used Delta Lake on Databricks to manage slowly changing dimensions in policyholder datasets.
- Validated data lineage using Great Expectations and built reusable data validation modules.
- Orchestrated multi-step pipelines using Apache Airflow, ensuring dependency handling and retries.
- Implemented partitioning and bucketing in S3 and Parquet formats to enhance query efficiency.
- Supported GDPR and SOC 2 audits by documenting security controls and data flow diagrams.
- Championed knowledge transfer and onboarding for Flink and Glue pipeline frameworks.
- Presented results of fraud model enhancements to executive leadership and stakeholders.
- Integrated Snowflake for high-speed dashboard reporting and analytics on curated datasets.

Environment: AWS Kinesis, AWS Lambda, AWS Glue, AWS SageMaker, Amazon Athena, S3, IAM, CloudWatch, EMR, Apache Flink, Apache Airflow, Databricks, Delta Lake, Great Expectations, Python, Spark, Terraform, JSON, Parquet, SOC 2, RBAC, JIRA

Elevance Health **MA, USA**
Big Data Engineer **Jul 2018 – Dec 2019**

- Designed and implemented Spark-based risk scoring pipelines for chronic disease prediction.
- Developed PySpark MLlib workflows to compute patient risk scores using claims and clinical data.
- Built scalable data lake architecture on Hadoop with HDFS, Hive, and Spark SQL.
- Enabled HIPAA-compliant data access by implementing encryption and role-based permissions.
- Automated data quality checks and ETL validation using Apache Airflow and custom validation scripts.
- Integrated external provider data and enriched risk profiles for improved care coordination.
- Tuned Spark jobs for parallel processing across 20+ node clusters, improving runtime by 40%.
- Created UDFs for business logic encapsulation in PySpark transformations.
- Developed schema evolution patterns using Hive and Avro for new provider data ingestion.

- Enabled self-service analytics by publishing structured datasets to Hive and building semantic views.
- Worked closely with actuaries and care teams to align model features with clinical priorities.
- Documented data ingestion flows, processing logic, and data dictionary in Confluence.
- Processed millions of claims per day using Spark Streaming and batch pipelines.
- Built resilient data pipelines to support data science experimentation on historical EHRs.
- Collaborated in Agile sprints, delivering iterative improvements to data reliability and model features.

Environment: Apache Spark, PySpark, Spark MLlib, Hadoop, Hive, HDFS, Spark SQL, Apache Airflow, Avro, JSON, Hive UDFs, Shell Scripting, Data Vault 2.0, Git, SQL, Python, Confluence, HIPAA

M&T Bank **NY, USA**
BI Engineer **Jan 2017 – Jun 2018**

- Built BI reporting solutions for regulatory compliance using SSRS and Power BI.
- Automated generation of Dodd-Frank and SOX compliance reports across multiple business units.
- Modeled reporting schemas in SQL Server using star schema and snowflake design.
- Designed interactive dashboards in Tableau for executive insights on customer transaction patterns.
- Enabled data-driven fraud detection by surfacing anomalous KPIs in near real-time.
- Collaborated with audit and compliance teams to define validation rules and reporting thresholds.
- Implemented version control and code review workflows using Git and Bitbucket.
- Tuned SQL queries and indexes to improve report refresh times and query latency.
- Developed ELT workflows to stage and transform data into reporting layers.
- Built KPI monitoring dashboards using Power BI service and scheduled data refreshes.
- Documented data sources, metric definitions, and report logic in Confluence.
- Partnered with infrastructure teams to optimize SQL Server memory and storage configurations.
- Conducted performance and usage audits of dashboards to prioritize enhancements.
- Integrated audit log data and access logs into BI reports for transparency and governance.
- Supported Tableau and Power BI onboarding for business users, including training materials.

Environment: SQL Server, SSRS, Power BI, Tableau, Git, Bitbucket, SQL, Python, Star Schema, Snowflake Schema, ELT, Agile, JIRA, Confluence, Compliance (SOX)

Cox Communications **GA, USA**
ETL Developer **May 2015 – Dec 2016**

- Developed ETL workflows using SSIS to consolidate customer data from Oracle and SQL Server.
- Designed and implemented a customer 360 data mart supporting marketing and service teams.
- Migrated historical billing data to new platforms, ensuring zero data loss and schema consistency.
- Built data validation scripts in Python and PL/SQL to compare source vs. target post migration.
- Maintained and tuned legacy batch jobs for nightly transformations and reporting.
- Developed audit and reconciliation checks to ensure data quality in ETL workflows.
- Integrated JIRA with deployment pipelines for defect tracking and change management.
- Automated file movement and ETL triggers using shell scripting in Linux.
- Modeled reporting layers using star schema for customer churn and satisfaction metrics.
- Collaborated with QA and business teams to perform UAT on newly migrated datasets.
- Documented ETL logic, mappings, and job runbooks for operational readiness.

- Used SSRS to create internal reports and logs for data validation tracking.
- Developed reusable SSIS templates to accelerate ETL for new source integrations.
- Implemented RBAC policies in SQL Server for data access control.
- Conducted root cause analysis of data mismatches and improved field-level mapping accuracy.

Environment: SQL Server, Oracle PL/SQL, SSIS, SSRS, Shell Scripting (Bash), PowerShell, Python, JIRA, Star Schema, RBAC, XML, CSV, Batch Processing, Legacy Systems